



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2026-2027

Vision: Dream of where you want.

Mission: Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords:

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Harshal Mohadikar
(09/07/2026)

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2026-27 (ODD)	Course Name	Lab: Generative AI
Semester	7	Course Code	23ADS1706
Roll No	41	Name of Student	Krishna Mishra

Practical Number	1
Course Outcome	CO1: Explain the fundamentals of Artificial Intelligence, Neural Networks, Natural Language Processing, and Transfer Learning used in Generative AI applications. CO 2: Apply Generative AI models such as GANs, VAEs, Transformers, BERT, GPT, and T5 to solve real-world AI and NLP problems. CO 3: Analyze and optimize prompts using prompt engineering techniques for improving the performance of Large Language Models in different application domains. CO 4: Design and develop Generative AI solutions by integrating Foundation Models, Retrieval-Augmented Generation (RAG), tokenization, fine-tuning, and chunking techniques for domain-specific applications.
Aim	To perform text preprocessing with NLP Techniques.
Problem Definition	The objective is to clean and normalize raw textual data to make it suitable for machine learning and NLP models. Raw text often contains irrelevant noise such as uppercase letters, punctuation marks, digits, URLs, and stopwords (e.g., "the," "is"), which do not contribute meaningful semantic information and can degrade model performance. This practical involves applying a sequential preprocessing pipeline—lowercasing, removing punctuation, removing numbers, and removing stopwords—on product review samples and a suspicious email to demonstrate the transformation of unstructured text into a cleaner format ready for downstream tasks like sentiment analysis and spam detection.
Theory (100 words)	Text preprocessing is a foundational step in Natural Language Processing (NLP) that transforms raw, unstructured text into a standardized format. <i>Lowercasing</i> ensures case-insensitivity, preventing the model from treating "Love" and "love" as distinct features. <i>Punctuation removal</i> eliminates non-alphanumeric characters that add noise. <i>Digit removal</i> strips numerical values that often do not generalize well across different contexts. <i>Stopword removal</i> filters out frequently occurring but



	semantically weak words (e.g., "and," "the") using NLTK's pre-built corpus, reducing the feature space and focusing the model on meaningful content tokens. Tokenization splits text into individual words, enabling these filtering operations. These techniques collectively reduce dimensionality and enhance algorithmic performance.
Procedure and Execution (100 Words)	Algorithm: 1. Imported required libraries: string, re, and NLTK modules (including stopwords and word_tokenize), ensuring necessary corpora were downloaded. 2. Defined a sample dataset containing four product reviews with uppercase, emoticons, numbers, and exclamation marks. 3. Applied a sequential preprocessing pipeline: converted text to lowercase using .lower(), removed punctuation using str.maketrans(), eliminated digits using re.sub(r'\d+', ''), and removed stopwords by tokenizing the text, filtering against NLTK's stopword set, and rejoining the remaining tokens. 4. Repeated the same pipeline on a sample email designed for spam detection, demonstrating the process on a more realistic use case. 5. Printed outputs after each stage to visualize the transformation. Code: <pre>sample_dataset = "1.I absolutely LOVE this phone!! Battery lasts 2 days. 2.Worst Product Ever :(Money Wasted !! 3.Delivery was quick, but the packaging was damaged :((4.Excellent camera quality and amazing performance !!"</pre> # 1.Convert text to lowercase <pre>sample_dataset = sample_dataset.lower() print(sample_dataset)</pre> 1.i absolutely love this phone!! battery lasts 2 days. 2.worst product ever :(money wasted !! 3.delivery was quick, but the packaging was damaged :((4.excellent camera quality and amazing performance !!" # 2.Remove Punctuation <pre>import string sample_dataset = sample_dataset.translate(str.maketrans("", "", string.punctuation)) print(sample_dataset)</pre> 1i absolutely love this phone battery lasts 2 days 2worst product ever money wasted 3delivery was quick but the packaging was damaged 4excellent camera quality and amazing performance # 3.Remove Numbers <pre>import re sample_dataset = re.sub(r'\d+', "", sample_dataset) print(sample_dataset)</pre> i absolutely love this phone battery lasts days worst product ever



money wasted delivery was quick but the packaging was damaged
excellent camera quality and amazing performance

4.Remove StopWords

```
import nltk
nltk.download('stopwords') # Download stopwords corpus
nltk.download('punkt') # Download punkt tokenizer for word_tokenize
nltk.download('punkt_tab') # Download punkt_tab as specifically
requested by error
```

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
stop_words = set(stopwords.words('english'))
words = word_tokenize(sample_dataset) # Tokenize the dataset
```

```
# Filter out stopwords
filtered_words = [word for word in words if word.lower() not in
stop_words]
sample_dataset = ''.join(filtered_words)
print(sample_dataset)
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
absolutely love phone battery lasts days worst product ever money
wasted delivery quick packaging damaged excellent camera quality
amazing performance
Apply Preprocessing to a Sample Email for Spam Detection
```

5. Define a sample email

```
sample_email = "Subject: Urgent: Your account has been
compromised! Click this link now: http://malicious.com/phish123 Get
FREE money! Limited time offer! Call 1-800-spam-now. "
print(f"Original Email:\n{sample_email}")
```

Original Email:

Subject: Urgent: Your account has been compromised! Click this link
now: http://malicious.com/phish123 Get FREE money! Limited time
offer! Call 1-800-spam-now.

6. Convert email to lowercase

```
sample_email = sample_email.lower()
print(f"\nAfter Lowercasing:\n{sample_email}")
```

After Lowercasing:

subject: urgent: your account has been compromised! click this link
now: http://malicious.com/phish123 get free money! limited time
offer! call 1-800-spam-now.

```
import string
```

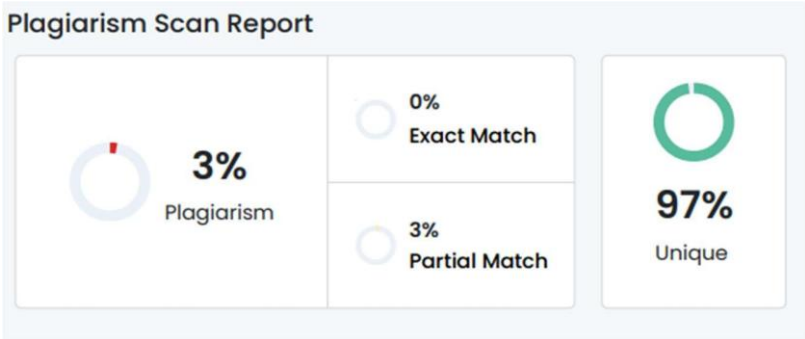
7. Remove Punctuation from email

```
sample_email = sample_email.translate(str.maketrans("", "",
string.punctuation))
print(f"\nAfter Punctuation Removal:\n{sample_email}")
```



	After Punctuation Removal: subject urgent your account has been compromised click this link now httpmaliciouscomphish123 get free money limited time offer call 1800spamnow import re # 8. Remove Numbers from email sample_email = re.sub(r'\d+', '', sample_email) print(f'\nAfter Number Removal:\n{sample_email}') After Number Removal: subject urgent your account has been compromised click this link now httpmaliciouscomphish get free money limited time offer call spamnow import nltk from nltk.corpus import stopwords from nltk.tokenize import word_tokenize # Ensure NLTK data is downloaded (if not already) nltk.download('stopwords', quiet=True) nltk.download('punkt', quiet=True) nltk.download('punkt_tab', quiet=True) # 9. Remove StopWords from email stop_words = set(stopwords.words('english')) email_words = word_tokenize(sample_email) # Tokenize the email # Filter out stopwords filtered_email_words = [word for word in email_words if word.lower() not in stop_words] sample_email_processed = ' '.join(filtered_email_words) print(f'\nAfter Stopword Removal:\n{sample_email_processed}') After Stopword Removal: subject urgent account compromised click link httpmaliciouscomphish get free money limited time offer call spamnow
Output Analysis	The preprocessing pipeline effectively stripped all noise from the raw texts. In the product reviews, sentiment-bearing keywords such as "love," "excellent," and "amazing" were retained while connectors like "the" and "was" were removed. For the spam email, critical indicators like "urgent," "compromised," "free," and "call" remained, while the URL and numbers were successfully eliminated. The final outputs are clean, concise token sequences that preserve semantic meaning, proving the pipeline's effectiveness for reducing dimensionality before model training.
Link of student Github profile where lab assignment has been uploaded.	krish-na45/GenAI-Practicals



Plag Report (Similarity index < 12%)	
Conclusion	The text preprocessing implementation using Python and NLTK successfully transformed noisy raw text into a structured and meaningful format. These fundamental cleaning steps are essential for building efficient and accurate NLP models, as they ensure that algorithms focus on relevant content rather than irrelevant surface-level noise.
Date	09/07/2026